

Advances in Financial M

Lecture 5/

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What are we going to

- Bet Sizing
 - Using Predicted Probabilities
 - Averaging Bets and Size Discretization
- The Dangers of Backtesting
 - Mission Impossible: The Flawless Backtest
 - Even Flawless Backtests are Usually Wrong
 - Some General Recommendations
- Backtesting Methods
 - Walk Forward
 - Cross-Validation
 - Combinatorial Purged Cross-Validation
 - Backtesting on Synthetic Data

Bet Sizing

Using Predicted Probab

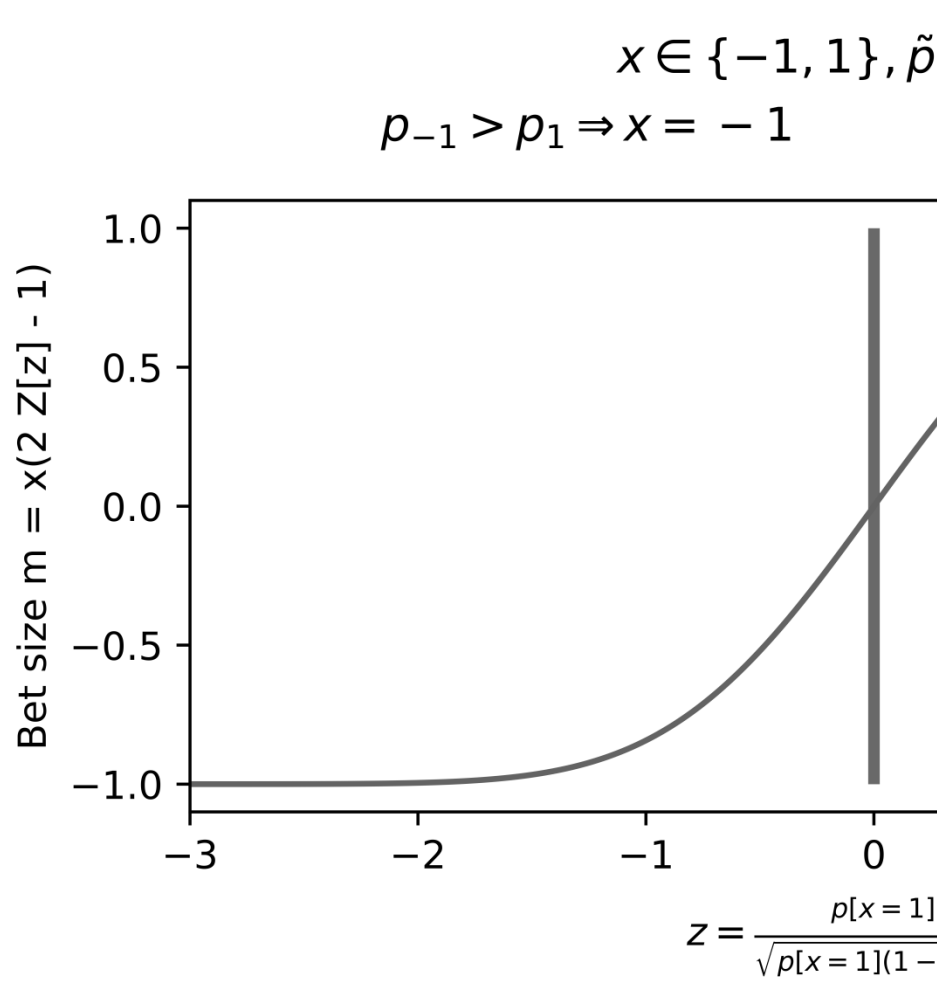
- Let us denote $p[x]$ the probability that label x is correct. We will use this predicted probability to derive the test statistic.
- For two possible outcomes, $x \in \{-1, 1\}$, we have the null hypothesis $H_0: p[x = 1] = \frac{1}{2}$.
- We compute the test statistic

$$z = \frac{p[x = 1] - \frac{1}{2}}{\sqrt{p[x = 1](1 - p[x = 1])}} = \frac{p[x = 1] - \frac{1}{2}}{\sqrt{p[x = 1]}}$$

with $z \in (-\infty, +\infty)$ and where Z represents the standard normal distribution.

- We derive the bet size as $m = 2Z[z] - 1$, where $Z[z]$ is the CDF of Z .

Using Predicted Probab



Using Predicted Probab

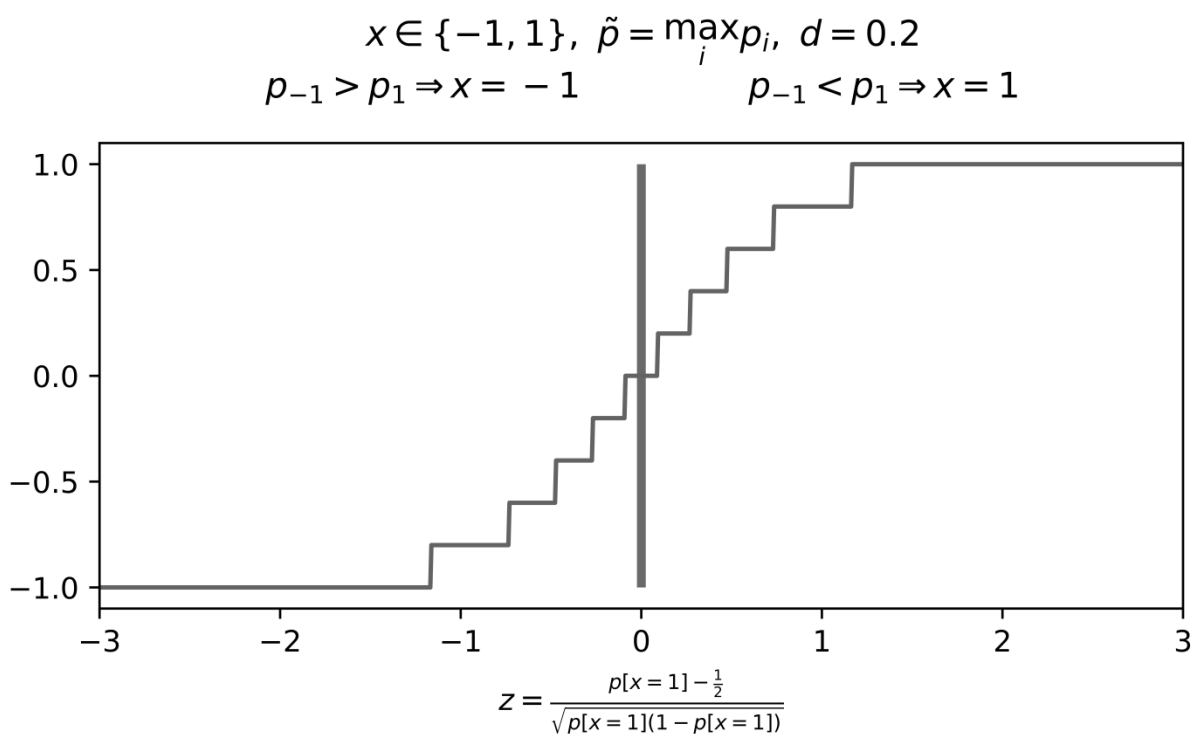
- For more than two possible outcomes, we follow a
- Let $X = \{-1, \dots, 0, \dots, 1\}$ be various labels associa
predicted label. In other words, the label is identifi
- For each label $i = 1, \dots, \|X\|$, we estimate a probab
- We define $\tilde{p} = \max_i \{p_i\}$ as the probability of x , and
- We compute the test statistic $z = \frac{\tilde{p} - \frac{1}{\|X\|}}{\sqrt{\tilde{p}(1-\tilde{p})}} \sim Z$, with
as $m = x \underbrace{(2Z[z] - 1)}_{\in [0,1]}$, where $m \in [-1,1]$ and $Z[z]$
(where the side is implied by x). Uncertainty is abs
likely.

Averaging B

- Every bet is associated with a holding period from the time it originated to the time the first barrier is hit (see 3).
- One possible approach is to override an existing bet, however, that is likely to lead to excessive trading.
- A more sensible approach is to average bets that are active at a given point in time.

Size Discretiza

- Averaging reduces some of the excess t
small trades will be triggered with every

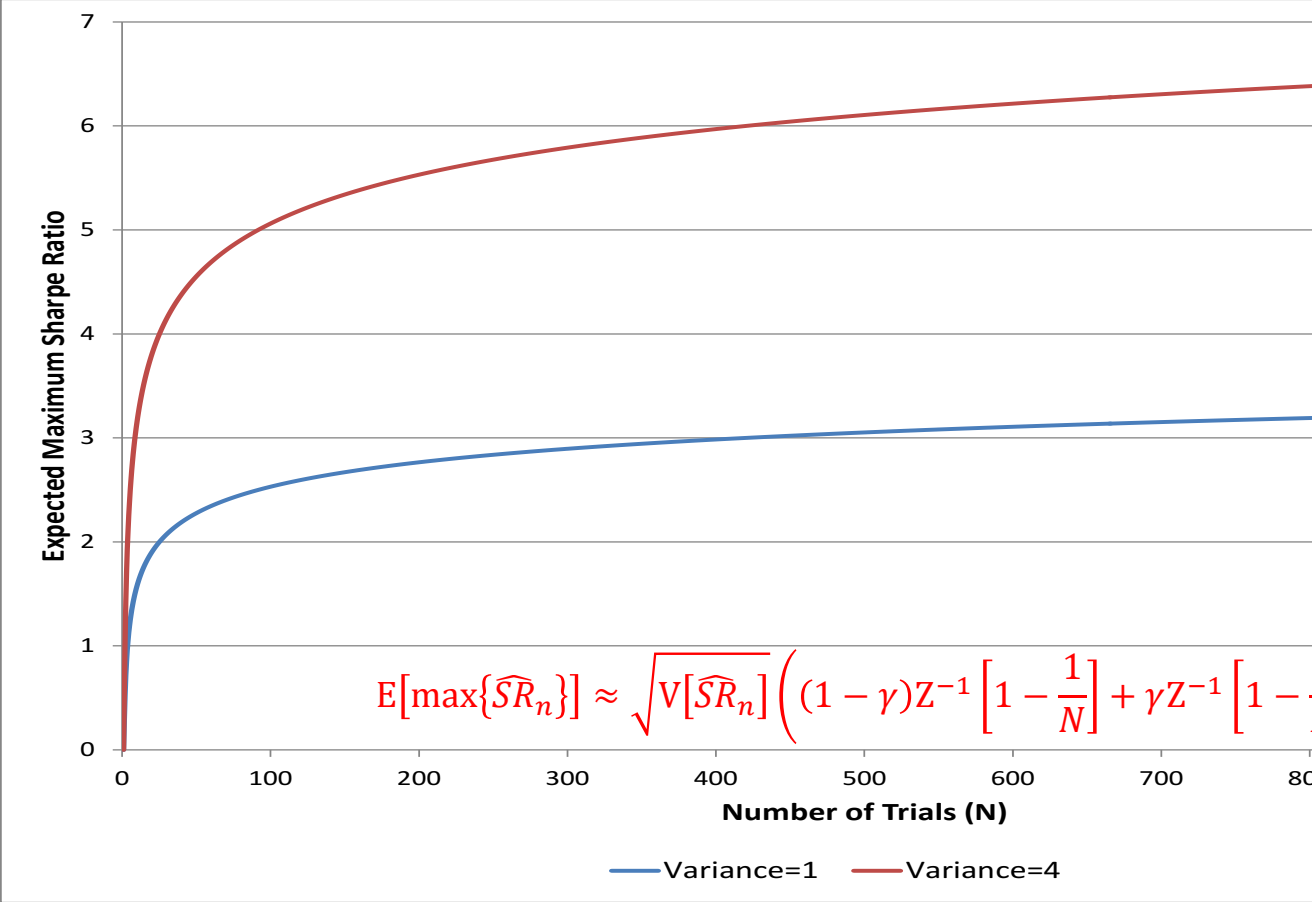


The Dangers of Ba

Mission Impossible: The F

- Common backtesting errors include:
 - **Survivorship bias:** Using as investment univers some companies went bankrupt and securities
 - **Look-ahead bias:** Using information that was n simulated decision would have been made. Be data point. Take into account release dates, dis corrections.
 - **Storytelling:** Making up a story *ex-post* to justifi
 - **Data mining and data snooping:** Training the n
 - **Transaction costs:** Simulating transaction costs certain about that cost would have been to int the actual trade).
 - **Outliers:** Basing a strategy on a few extreme o as observed in the past.
 - **Shorting:** Taking a short position on cash produ of lending and the amount available is general inventory, relative demand, etc.

Even Flawless Backtests a



This is a consequence of pure random behavior. We will find high S associated with this strategy class ($E[\widehat{SR}_n] = 0$).

Some General Recom

- While there is no easy way to prevent overfitting, we can reduce its presence.
- 1. **Develop models for entire asset classes** or investment universe (Chapter 8). Investors diversify, hence they do not make money only on security Y , no matter how apparently profitable, it is.
- 2. **Apply bagging** (Chapter 6) as a means to both prevent overfitting and reduce forecasting error. If bagging deteriorates the performance, increase the number of observations or outliers.
- 3. **Do not backtest until all your research is complete** (Chapter 10).
- 4. **Record every backtest conducted on a dataset** so that the results are not estimated on the final selected result (see Bailey, Borwein, 2014), and the Sharpe ratio may be properly deflated by the number of tests (de Prado [2014b]).
- 5. **Simulate scenarios rather than history** (Chapter 12). A strategy can be easily overfit. History is just the random path that was realized. Your strategy should be profitable under a wide range of scenarios, not just the historical path. It is harder to overfit the outcome of thousands of simulations.
- 6. **Do not research under the influence of a backtest.** If the backtest is profitable, start from scratch. Resist the temptation of reusing those results.

Backtesting M

Walk Forward

- WF is a historical simulation of how the strategy would have performed in the past.
 - Each strategy decision is based on observations up to the current time.
- WF enjoys two key advantages:
 - WF has a clear historical interpretation. Its performance is measured as if the strategy were paper trading.
 - History is a filtration; hence, using trailing data is out-of-sample (no leakage), as long as purging has been implemented (see Chapter 7, section 7.4.1).
- A common mistake found in WF backtests is **purging**, where `t1.index` falls within the testing set (see Chapter 3).
- **Embargoing** is not needed in WF backtests as the strategy's performance predates the testing set.

Walk Forward

- WF suffers from three major disadvantages
 - A single scenario is tested (the historical path) (Bailey et al. [2014]).
 - WF is not necessarily representative of future outcomes, as it is biased by the particular sequence of data points used. It is possible to overfit a walk-forward backtest as to overfit to the fact that changing the sequence of observations and outcomes is evidence of that overfitting.
 - The initial decisions are made on a smaller dataset. If a warm-up period is set, most of the information is lost in the portion of the decisions.

Cross-Validation

- The goal of backtesting through cross-validation is not to measure accurate performance, but to infer future performance on new sample scenarios. For each period of the backtest, we assume the existence of a classifier that knew everything except for the test data.
- **Advantages**
 1. The test is not the result of a particular sequence of events. If we are walking forward, other outcomes result from walking forward. If we are training data from before and after the testing sequence, we get to an entirely different CV result.
 2. Every decision is made on sets of equal size. This is true for all periods, in terms of the amount of information used.
 3. Every observation is part of one and only one test set, thereby achieving the longest possible out-of-sample performance.

Cross-Validation

- **Disadvantages**

1. Like WF, a single backtest path is simulated (i.e., only one). There is one and only one forecast.
2. CV has no clear historical interpretation. It asks the question: how would the strategy have performed in the *future* under various stress scenarios?
3. Because the training set does not trail the test set, extreme care must be taken to avoid leakage from the test set into the training set. See Chapter 7 for a discussion of how cross-validation can help prevent informational leakage in machine learning.

Combinatorial Purged Cross

- CPCV addresses the main drawback of the WF and CV methods: a single path.
- Consider T observations partitioned into N groups without replacement. If the first group is of size $\lfloor T/N \rfloor$, the N th group is of size $T - \lfloor T/N \rfloor(N - 1)$, and if we select a testing set of size k groups, the number of possible training sets is

$$\binom{N}{N-k} = \frac{\prod_{i=0}^{k-1} (N-i)}{k!}$$

- Since each combination involves k tested groups, the total number of combinations is $\binom{N}{k}$. Since we have computed all possible combinations, these tests are repeated across all N (each group belongs to the same number of training sets). That is, from k -sized testing sets on N groups we can backtest

$$\varphi[N, k] = \frac{k}{N} \binom{N}{N-k} = \frac{k}{N} \binom{N}{k}$$

Combinatorial Purged Cross

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	Paths
G1	x	x	x	x	x											5
G2	x					x	x	x	x							5
G3		x				x				x	x	x				5
G4			x				x			x			x	x		5
G5				x				x			x		x		x	5
G6					x				x			x		x	x	5

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	Paths
G1	1	2	3	4	5											5
G2	1					2	3	4	5							5
G3		1				2				3	4	5				5
G4			1				2			3			4	5		5
G5				1				2			3		4		5	5
G6					1				2			3		4	5	5

Composition
 $\binom{6}{4} = 15$ spl
marks with a
and leaves un
Each group fo
train/test spl

Assignment o
example, pat
(G1, S1), (G2
Path 2 is the
(G2, S6), (G3

- In the example above we have generated only 5 paths, however C a sufficiently long series. The number of paths $\varphi[N, k]$ increases w
- A key advantage of CPCV is that it allows us to derive a distribution overfit) Sharpe ratio estimate.

Backtesting on Synthetic Data

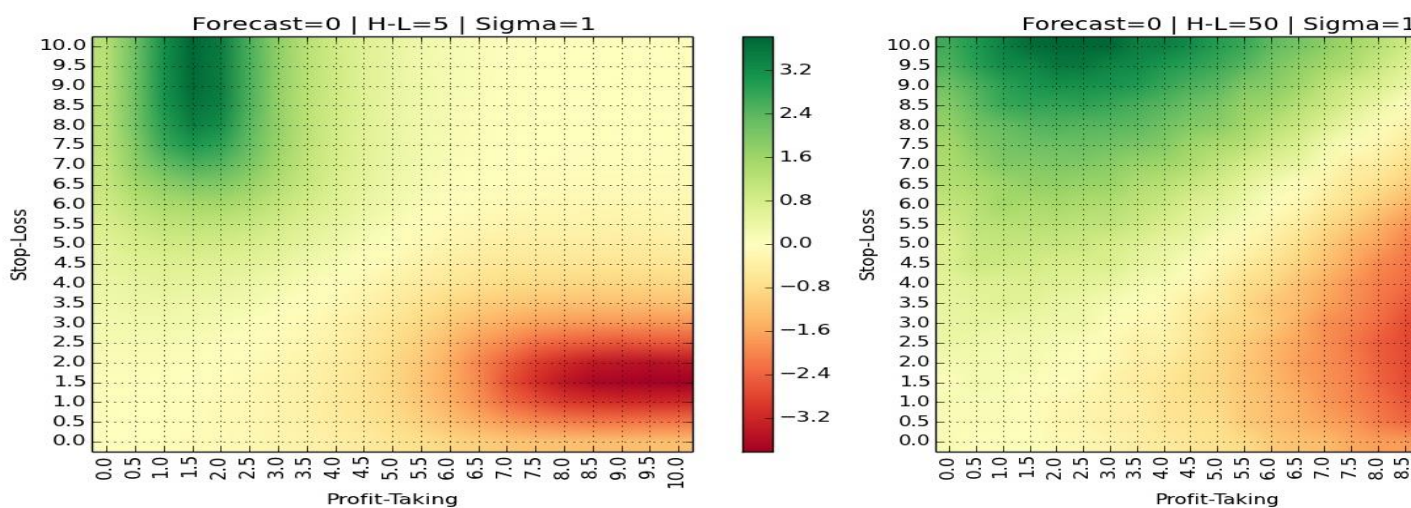
- A key problem in finance is that datasets are only available for the past.
- Alternatively, we could generate synthetic data for a particular phenomenon that we wish to study.
 1. Estimate the input parameters $\{\sigma, \varphi\}$ from the historical data using the formula
$$P_{i,t} = E_0[P_{i,T_i}] + \varphi(P_{i,t-1} - E_0[P_{i,t-1}]) + \sigma \epsilon_t$$
 2. Construct a mesh of stop-loss and profit-taking pairs.
 3. Generate a large number of paths (e.g., 100,000) for each node.
 4. Apply the 100,000 paths generated in Step 3 on each node of Step 2.
 - For each node, we apply the stop-loss and profit-taking logic.
 - For each node we compute the Sharpe ratio associated with the strategy.

Backtesting on Synthetic

5. Compute the solution:

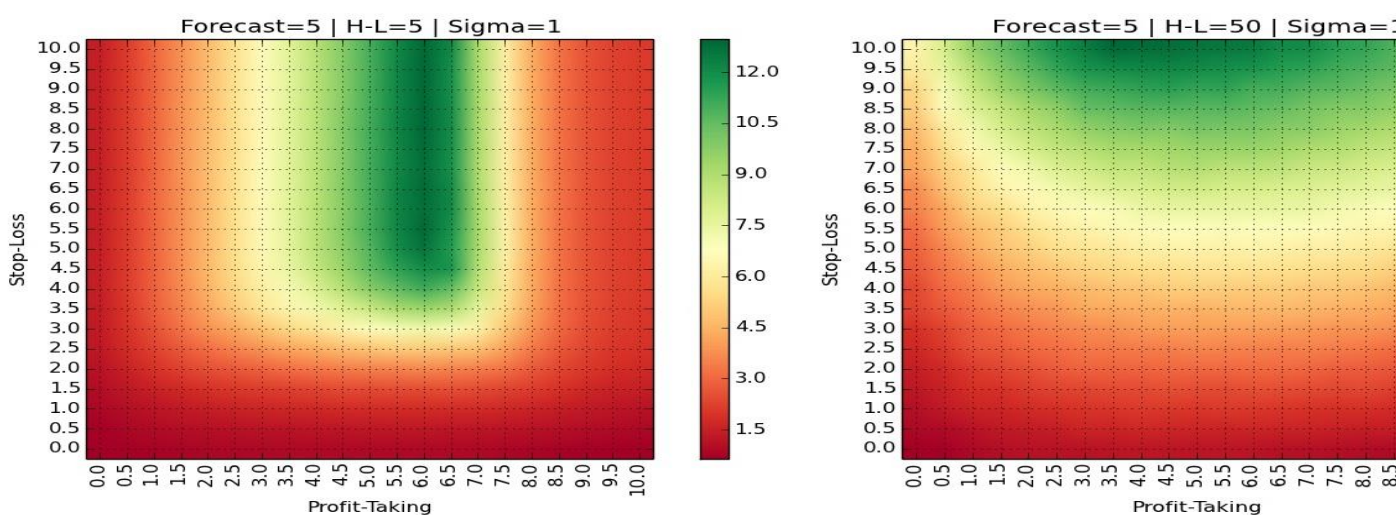
- We determine the pair $(\underline{\pi}_i, \overline{\pi}_i)$ within the mesh of input parameters $\{\hat{\sigma}, \hat{\phi}\}$ and the observed initial conditions.
- If strategy S provides a profit target $\overline{\pi}_i$ for a particular piece of information in conjunction with the results in Step 4.
- If the trader has a maximum stop-loss $\underline{\pi}_i$ imposed on the strategy, that information in conjunction with the results in Step 4, taking $\overline{\pi}_i$ within the range of stop-losses $[0, \underline{\pi}_i]$.

Cases with Zero Long-R



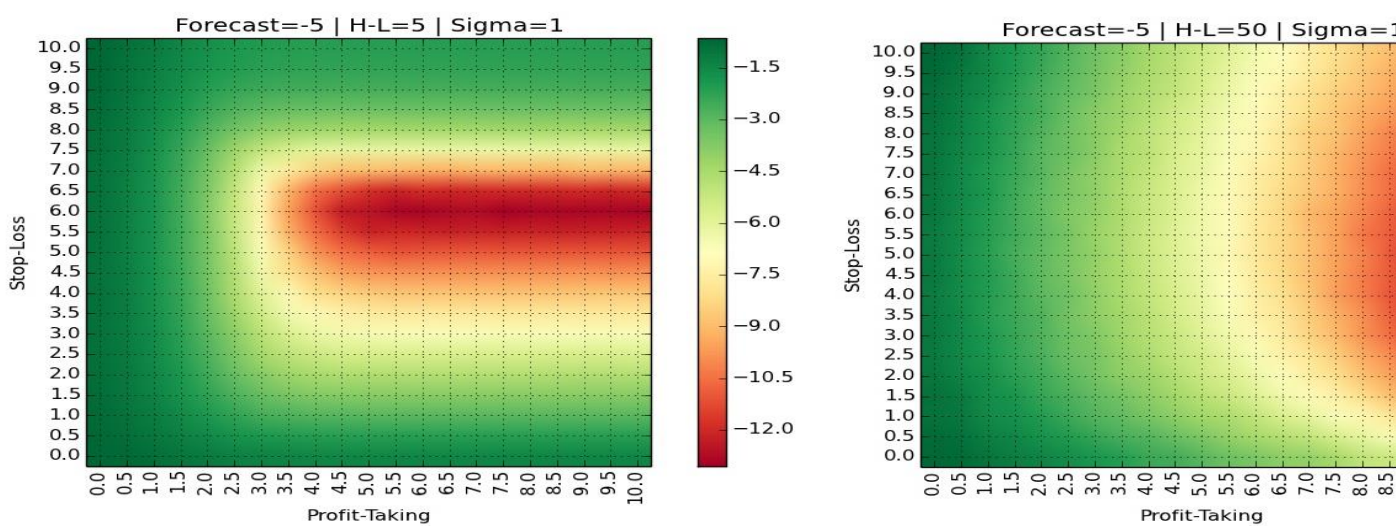
- From left to right, these figures show the Sharpe ratios for parameters $\{\mu, \tau, \sigma\} = \{0, 100, 1\}$.
- For small half-life, performance is maximized in a narrow range of losses: The optimal trading rule is to hold an inventory long enough experiencing 5 or 7-fold losses.
- **This is in fact what many market-makers do in practice, and is commonly described in Easley et al. [2011].**
- The worst possible trading rule in this setting would be to combine a situation that market-makers avoid in practice.

Cases with Positive Long-

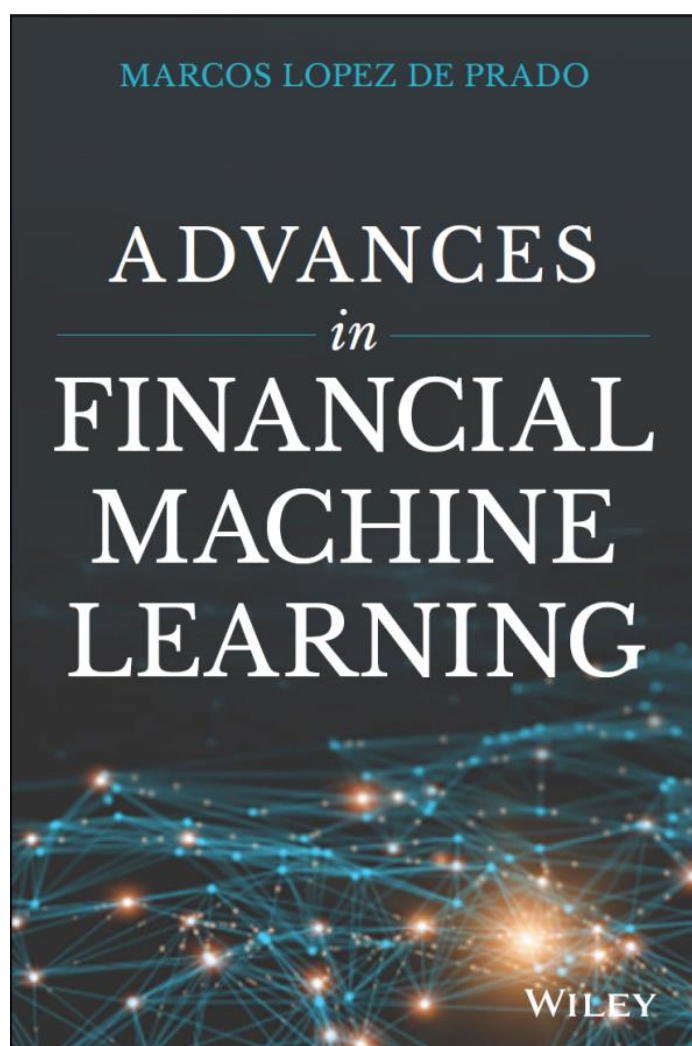


- From left to right, these figures show the Sharpe ratios for parameters $\{5, 50, 1\}$, $\{\mu, \tau, \sigma\} = \{5, 100, 1\}$.
- Because positions tend to make money, the optimal profit-taking is high.
- As we increase the half-life, the range of optimal profit-taking widens, shaping the initial rectangular area closer to a square and then a square.
- Again, a larger half-life brings the process closer to a random walk.

Cases with Negative Long



- From left to right, these figures show the Sharpe ratios for parameters $\{-5, 50, 1\}$, $\{\mu, \tau, \sigma\} = \{-5, 100, 1\}$.
- Results appear to be rotated complementaries of what we obtained (negatives).
- The reason is, that the profit in the previous figures translates into image of the other, just as a gambler's loss is the house's gain.



For Additional

The first wave of quantitative optimization. Machine Learning is an essential aspect of finance. López de Prado's book is essential for readers who want to avoid being replaced by it.

— Prof. **Campbell Harvey**, Distinguished Professor, American Finance Association

Financial problems require vast amounts of data. López de Prado's book is the only one that shows how machine learning tools fail without human intervention. It is one to provide practical solutions for financial managers. Everyone who wants to succeed should read this book.

— Prof. **Frank Fabozzi**, EDHEC Business School, Portfolio Management.

THANKS FOR YOUR A

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